UPI Fraud Stress Lab

Can an attack campaign reach its target before a detector stops it?

A synthetic campaign-level research laboratory

PROJECT GUIDE

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Final project branch: prototype/rl-academic-complete

Offline simulation only. No real payments. Gate C remains HOLD.

Why measure a whole campaign?

A transaction label is not the same as an intervention outcome.

ALLOW	→	FLAG	→	BLOCK
Transfer can proceed.		Alert raised; transfer may proceed.		Simulator stops the campaign.

Measure target completion, first alert, first block and gross simulated debit.

Source: prototype source and retained records

How the prototype works



Feedback: detector → adaptive attacker (public decisions only).

Two saved detectors

XGBoost and Random Forest
Reused artifacts; no retraining required for the demo.

Four workload families

High-value burst · Slow drain
Gradual escalation · Beneficiary hopping

Source: prototype source and retained records

Compare four attacker controls

Static	Family rules that do not respond to detector feedback.
Rule-adaptive	Hand-written rules respond to observed decisions.
Trained RL	Tabular reinforcement learning (RL), trained before evaluation.
Zero-Q control	Same RL action vocabulary without learned Q values.

RL: at most 64 states and 16 amount / pacing / recipient actions.

Source: prototype source and retained records

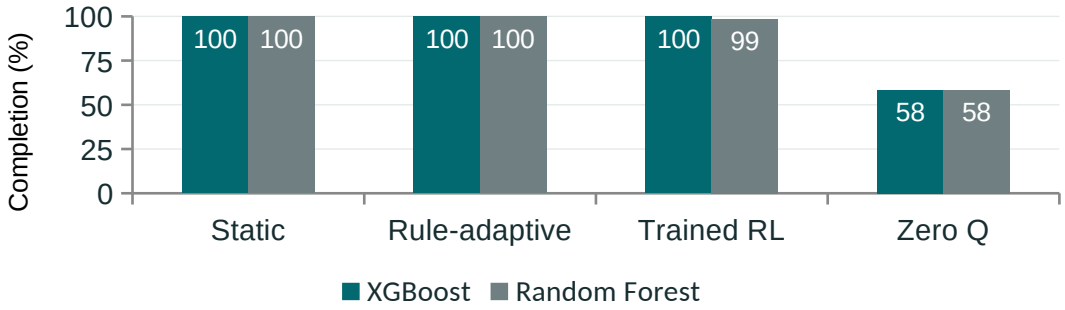
What the retained records show

1,152
attack campaigns

4,189
evaluated transactions

3
evaluation seeds

High-value-burst subset: campaign completion



All campaigns (pooled)
1,077 / 1,152 = 93.490%
reached their target.

Not detector accuracy
or proof of RL superiority.

Chart: high-value burst only.
72 per arm / detector; 3 seeds.
Seed SDs in notes; confounded.

Source: prototype source and retained records

Terminal results: what do they mean?

Recorded short demo: 16 campaigns, 4 training episodes; 1 campaign per cell.

```
xgboost known static success=1.000 exposure=46791.56
xgboost known rl success=1.000 exposure=29789.42
random_forest known rl success=1.000 exposure=29789.42
```

success=1.000	That configuration reached its target in 1 of 1 campaigns.
exposure	Gross simulated source debit in INR, not actual victim loss.
known	An attacker-training workload label, not proven detector holdout.
Interpretation	A tiny demo proves execution, not RL superiority or money saved.

Source: saved short-run stdout and interpretation

Run the prototype in a terminal

From a clone of prototype/rl-academic-complete; Linux / WSL2, Python 3.11.

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-milestone1.lock
python -B -s run_prototype.py \
  --detector both --attacker all --seeds 101 \
  --episodes 2 --campaigns 1 --output demo-faculty-01
```

Inspect campaigns.json, transactions.csv, summary.csv and comparison.svg.

Source: prototype source and retained records

From master to our final prototype

Area	master: earlier baseline	prototype: selected FINAL
Learning	Static + rule policies	Adds Q-learning + zero-Q control
Run / reload	Scenario runner	RL runner + saved-policy evaluation
Evidence	Scenario summaries	Traces, checkpoints, manifests, plots
Shared core	Two detectors + synthetic data	Same model/data bytes; architecture retained

Engineering contribution, not a controlled performance comparison.

Source: verified master-to-prototype code comparison